Round Robin Scheduling – 1-Minute Reading

Round Robin is a CPU scheduling algorithm designed to give every process a fair chance to use the CPU. Each process receives a fixed amount of CPU time called a **time quantum** or **time slice**.

Processes are organized in a **FIFO (First-In, First-Out) queue**. When a process uses up its time quantum without finishing, it moves to the back of the ready queue. If it finishes early, the CPU immediately moves to the next process.

Round Robin is commonly used in **time-sharing and multitasking operating systems** because it allows multiple processes to run responsively.

Switching the CPU from one process to another is called a **context switch**. If the time quantum is too small, frequent context switches can reduce system efficiency.

For example, if a process needs **5 milliseconds** of CPU time and the time quantum is **2 milliseconds**, it requires **3 CPU turns** to finish.

A practical application of Round Robin scheduling is managing multiple applications running on a computer, allowing each to receive CPU time without one process monopolizing the processor.